



Central Claim

Static harmful-prefix SFT can suppress known attacks in-distribution, but it does not learn a fully robust refusal basin over unsafe continuation states. Adaptive mining finds prefixes that remain vulnerable after SFT; mixed static+adaptive recovery improves robustness by raising refusal pressure in hidden-state trajectories.

- **Static SFT works ID**: base Llama-3.2-3B static-prefix ASR at $k = 10$ is **93.3%**; original augmented drops to **7.6%**.
- **Adaptive prefixes expose OOD**: same harmful prompt source, different prefilled continuation distribution.
- **Recovery must avoid forgetting**: adaptive-only repair improves mined failures but regresses on static prefixes.
- **Mechanism**: mixed recovery increases refusal margin at the prefix boundary on nearly every paired adaptive example.

Experimental Setup

We evaluate safety under **prefilled harmful continuations**. For each harmful request, the model receives the user prompt plus the first k tokens of an unsafe assistant response. A failure occurs when the model continues unsafe behavior rather than refusing or redirecting.

Prefix families

- **Static HEx-PHI**: fixed harmful assistant prefixes.
- **Adaptive mined**: model-generated prefixes kept only if they induce unsafe continuation after verification.
- **Cross-adaptive**: Qwen-mined prefixes evaluated on Llama and Llama-mined prefixes evaluated on Qwen.

Models

- Base Llama-3.2-3B and Qwen3-4B-Instruct.
- Original static-prefix augmented models.
- Adaptive-only recovery model.
- Mixed 50/50 static+adaptive recovery model.

Protocol and Caveats

Attack success is measured with the same keyword safety evaluator across runs. The strongest causal comparisons use matched prompt sets and paired prefixes. Cross-model results are interpreted as **partial transfer evidence**, not as a calibrated ranking of Qwen vs Llama robustness.

Conference-safe claim: static-prefix SFT leaves an exposed adaptive continuation-state distribution; mixed recovery partly repairs it without forgetting static safety.

1. Static Safety Does Not Imply Adaptive Safety

Static augmentation creates a strong in-distribution result, but adaptive mining identifies continuation states where the same model remains vulnerable.

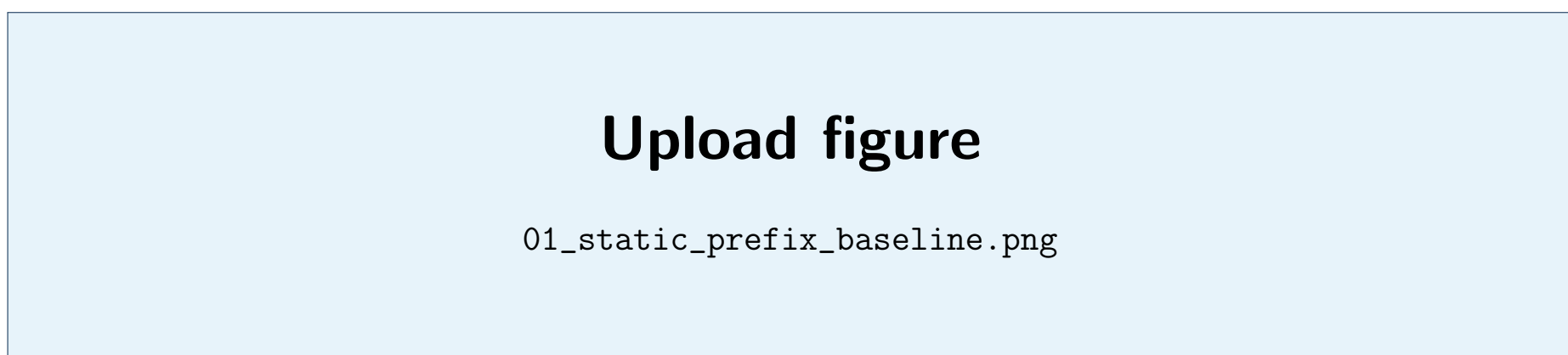


Figure: Static harmful-prefix augmentation sharply reduces ASR on the original prefix family.

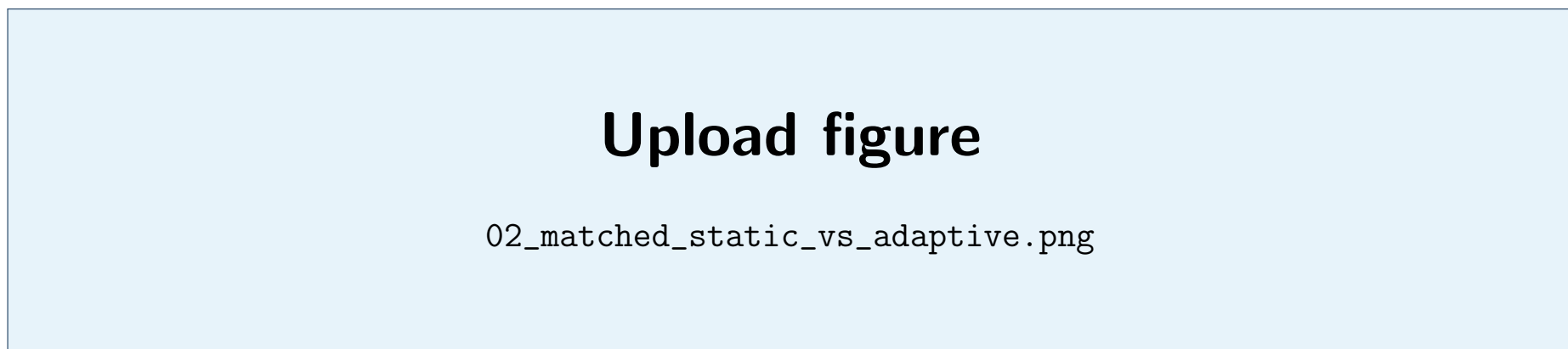


Figure: On matched HEx-PHI prompts, adaptive prefixes become harder than static prefixes at longer prefix lengths.

Condition	Prefix length	ASR
Base Llama, static	$k = 10$	93.3%
Original augmented, static	$k = 10$	7.6%
Original augmented, gen3 adaptive	$k = 40$	39.1%

2. Recovery Must Preserve the Original Basin

Adaptive-only recovery solves one part of the problem but regresses on static prefixes. Mixed recovery is the cleaner intervention because it keeps old robustness while adding robustness to mined failures.

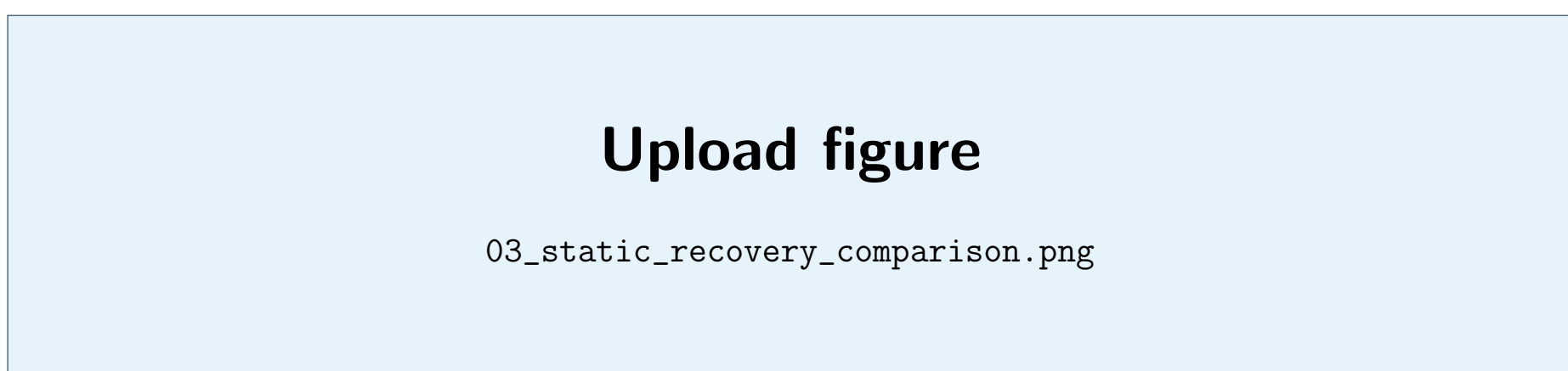
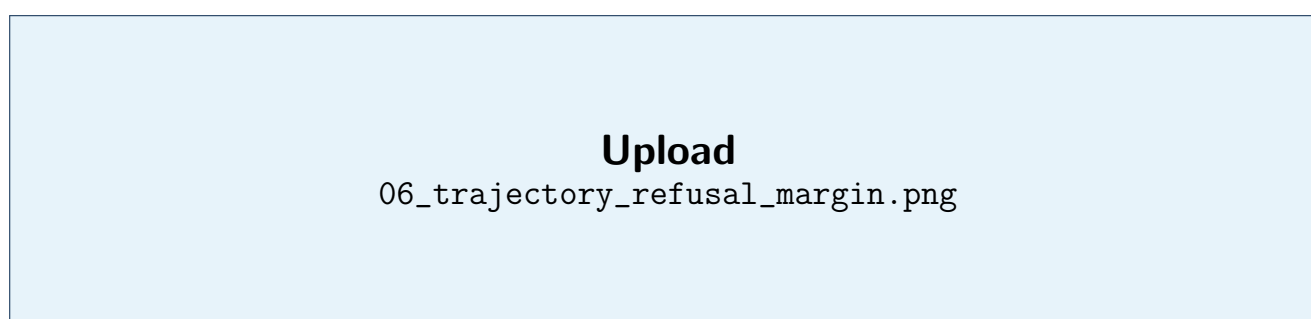


Figure: Adaptive-only recovery forgets static-prefix robustness; mixed recovery keeps static ASR near zero.

Model	Static $k = 10$	Static $k = 40$
Adaptive-only recovery	24.2%	30.3%
Mixed 50/50 recovery	2.1%	4.8%

5. Mechanism: Mixed Recovery Raises Refusal Pressure

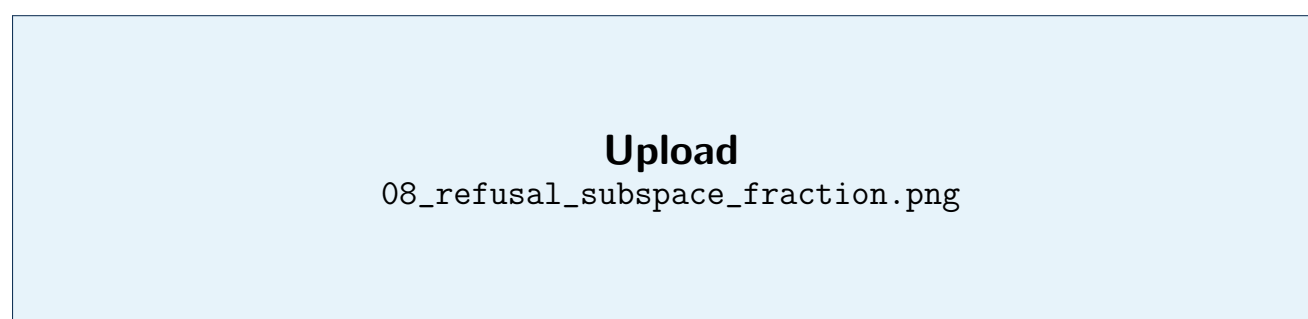
Hidden-state trajectory analysis suggests that mixed recovery does not simply memorize adaptive prefixes or collapse them into the static distribution. Instead, it increases refusal pressure at the prefix boundary and during early response generation.



(a) Refusal margin increases on paired adaptive prefixes.



(b) Most paired examples move in the safer direction.



(c) Adaptive prefixes are not merely standard refusal states.

Figure: Trajectory diagnostics: mixed recovery changes refusal dynamics, not just behavioral labels.

Comparison	Prefix length	Step	Mean refusal-margin gain
Mixed vs original augmented	$k = 20$	0	+4.54, positive on 23/23
Mixed vs original augmented	$k = 40$	0	+4.77, positive on 22/23
Mixed vs original augmented	$k = 20$	16	+2.98, positive on 20/23
Mixed vs original augmented	$k = 40$	16	+2.83, positive on 17/23

3. Mixed Recovery Reduces Adaptive Failures

On a paired gen3 hard set, both adaptive-only and mixed recovery reduce mined-prefix failures. The difference is that mixed recovery does so without the static-prefix regression.

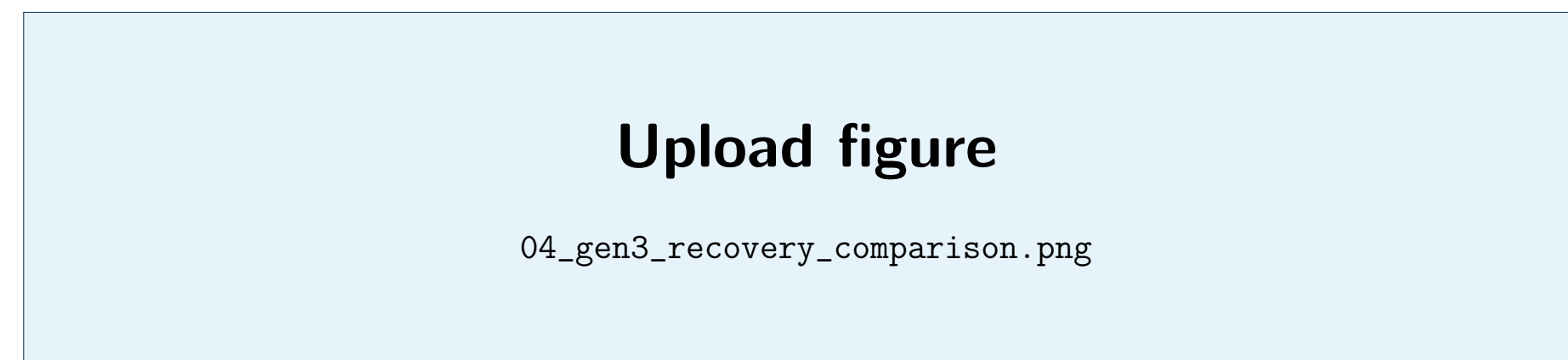


Figure: On gen3 adaptive prefixes, mixed recovery cuts $k = 40$ ASR from 39.1% to 13.0%.

Model	Gen3 $k = 20$	Gen3 $k = 40$
Original augmented	26.1%	39.1%
Adaptive-only recovery	8.7%	8.7%
Mixed 50/50 recovery	8.7%	13.0%

Mixed recovery is slightly weaker than adaptive-only on the adaptive hard set, but far stronger on static prefixes.

4. Adaptive Prefixes Are Measurably Distinct

A lightweight classifier separates matched static vs adaptive prefixes, supporting the claim that adaptive mining changes the prefix distribution rather than merely adding more unsafe examples.

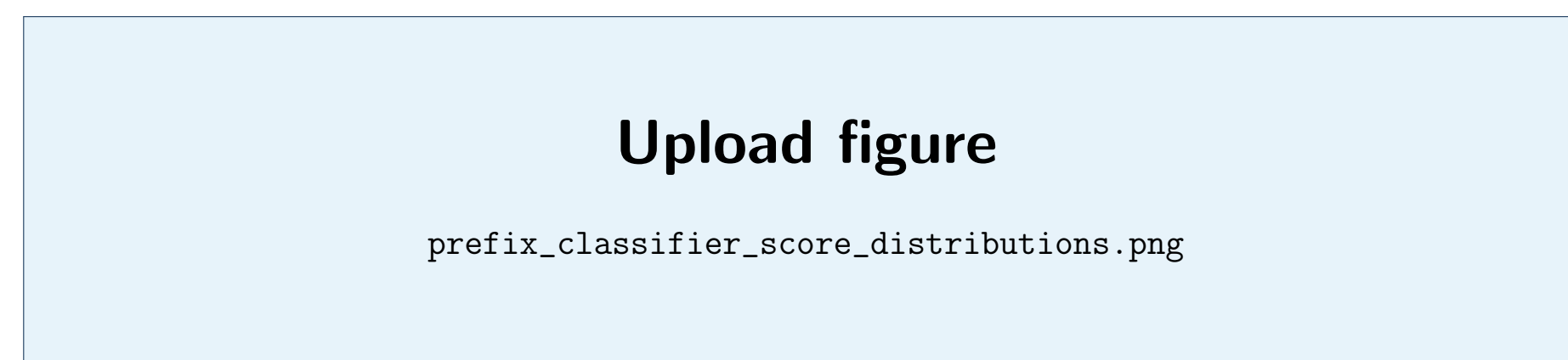


Figure: Static vs adaptive prefixes are separable by feature classifiers.

Feature set	CV AUC	Gen3 AUC
All features	1.000	1.000
No special tokens	0.993	0.989
Semantic/overlap only	0.798	0.586

Interpretation: the shift is mostly generation-state/format structure, not just harmful topic semantics.

6. Controlled Cross-Adaptive Evaluation

We evaluate adaptive prefixes mined from each original static-augmented model on both original static-augmented targets. This tests whether mined unsafe continuation states are purely model-specific or partially transferable.

Target	Prefix source	$k = 40$	Full prefix
Llama static-aug	Llama-mined	21.5%	30.7%
Llama static-aug	Qwen-mined	14.8%	21.0%
Qwen static-aug	Qwen-mined	48.8%	50.1%
Qwen static-aug	Llama-mined	55.8%	57.7%

Interpretation: adaptive vulnerabilities transfer, but asymmetrically. Treat this as evidence of partial transfer, not a calibrated architecture ranking.

Current original mined pools differ in retention and formatting; a protocol-matched Llama mining rerun is underway with Qwen-matched constraints.

7. Refusal-Subspace Caution

Adaptive prefixes are not simply ordinary refusal states. Their hidden-state deltas project less into the explicit refusal subspace than direct refusal prefixes, suggesting adaptive mining exploits broader continuation-state geometry.

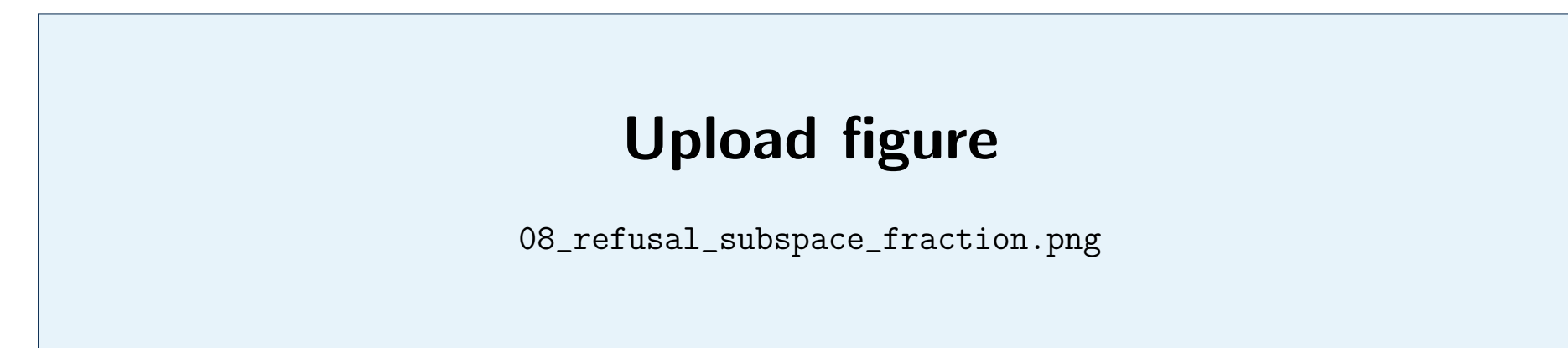


Figure: Adaptive prefixes occupy less of the explicit refusal subspace than refusal prefixes.

Prefix length	Adaptive frac.	Refusal frac.	Clean frac.
$k = 10$	0.335	0.645	0.395
$k = 20$	0.323	0.627	0.398
$k = 40$	0.350	0.512	0.397

Takeaways

- **Static-prefix SFT is real but narrow**: it sharply reduces ASR on the original static prefix family.
- **Adaptive mining reveals residual OOD vulnerability**: original augmented models still fail on mined continuation states.
- **Mixed recovery is the best trade-off**: it reduces adaptive failures while preserving static-prefix robustness.
- **Hidden states support the mechanism**: recovery raises refusal pressure at the prefix boundary, while adaptive prefixes remain geometrically distinct from simple refusal states.

Artifacts and Files to Upload

- Main artifacts: logs/final_adaptive_ood_package/
- Upload figures: 01_static_prefix_baseline.png, 02_matched_static_vs_adaptive.png, 03_static_recovery_comparison.png, 04_gen3_recovery_comparison.png
- Upload figures: prefix_classifier_score_distributions.png, 06_trajectory_refusal_margin.png, 07_trajectory-paired-signs.png, 08_refusal_subspace_fraction.png
- Controlled cross-adaptive table: logs/cross_adaptive_original_augmented_qwen4b_llama3b/cross_adaptive_summary.md